Blueprint for an Analytic Schwinger–Keldysh Kernel Pipeline

calc_mbcwc project

2026-05-20

1 Purpose

The goal is to define the mathematics that the Python pipeline should implement. The input is a weakly coupled finite-temperature model, specified by its quadratic kernel, interaction tensors, statistics, and index structure. The output is the leading large-time Schwinger–Keldysh ladder kernel and its largest positive growth exponent

$$\lambda_L = \max \operatorname{Re} \operatorname{spec} M, \tag{1}$$

where M is the linearized ladder evolution operator.

The immediate acceptance benchmark is Stanford's weakly coupled Hermitian matrix Φ^4 theory [1]. The general pipeline should reduce to the formulas in Sections 8 and 9 when that model is selected.

2 Model Input

Represent the Lorentzian action as

$$S[\phi] = \frac{1}{2}\int_x \phi_I(x) D^{-1}_{IJ}(\partial)\phi_J(x) - \sum_{\alpha\in\mathcal{V}} \frac{g_\alpha}{n_\alpha!}\int_x T^\alpha_{I_1\cdots I_{n_\alpha}} \phi_{I_1}(x)\cdots\phi_{I_{n_\alpha}}(x). \tag{2}$$

Here I, J are collective labels: species, matrix indices, internal indices, and spin/statistics labels. The parser should store:

- the quadratic inverse propagator $D^{-1}_{IJ}(k)$;
- interaction arity n_α, coupling g_α, and tensor T^α;
- index-contraction rules and large-N power counting;
- dispersion branches $E_a(\mathbf{p})$, either exact or from the poles of the retarded propagator.

3 Schwinger–Keldysh Doubling

For each bosonic field introduce contour fields ϕ_1, ϕ_2, or equivalently r/a fields

$$\phi_1 = \phi_r + \frac{1}{2}\phi_a, \qquad \phi_2 = \phi_r - \frac{1}{2}\phi_a. \tag{3}$$

The Schwinger–Keldysh action is

$$S_{\mathrm{SK}}[\phi_r, \phi_a] = S[\phi_1] - S[\phi_2]. \tag{4}$$

For a single monomial ϕ^n, the branch-difference identity is

$$\phi_1^n - \phi_2^n = \sum_{\substack{s=1 \\ s \text{ odd}}}^{n} \binom{n}{s} 2^{1-s} \phi_r^{n-s} \phi_a^s. \tag{5}$$

Thus every allowed SK interaction vertex contains an odd number of a-legs. This is a required implementation check:

$$S_{\mathrm{SK}}[\phi_r, \phi_a = 0] = 0. \tag{6}$$

With the convention in Eq. (2), differentiating the interaction part of $\mathrm{i}S_{\mathrm{SK}}$ gives the r/a-basis vertex

$$V^{\alpha}_{b_1\cdots b_n;I_1\cdots I_n} = -\mathrm{i}g_\alpha\, T^{\alpha}_{I_1\cdots I_n} C(b_1, \ldots, b_n), \qquad b_i \in \{r, a\}, \tag{7}$$

where

$$C(b_1, \ldots, b_n) = \begin{cases} 2^{1-s}, & s = \#\{i : b_i = a\} \text{ odd}, \\ 0, & s \text{ even}. \end{cases} \tag{8}$$

If T^α is not fully symmetrized, the implementation must sum over inequivalent cyclic/orderings specified by the model file.

4 Thermal Propagators

For each quasiparticle branch a, define

$$E_a(\mathbf{p}) = \sqrt{\mathbf{p}^2 + m_a^2} \quad \text{in the scalar benchmark,} \tag{9}$$

and use the retarded propagator

$$G_{\mathrm{R},a}(k) = \frac{\mathrm{i}}{(k^0 + \mathrm{i}0)^2 - E_a(\mathbf{k})^2 - \Pi_{\mathrm{R},a}(k)}. \tag{10}$$

Equivalently, in spectral form,

$$G_{\mathrm{R},a}(k) = \mathrm{i} \int \frac{\mathrm{d}\omega}{2\pi} \frac{\rho_a(\omega, \mathbf{k})}{k^0 - \omega + \mathrm{i}0}. \tag{11}$$

The half-thermal-circle Wightman function used for rungs is

$$\widetilde{G}_a(k) = \frac{\rho_a(k)}{2\sinh(\beta k^0/2)} \tag{12}$$

for bosons. For a free scalar,

$$\rho(k) = \frac{\pi}{E_{\mathbf{k}}} \left[\delta(k^0 - E_{\mathbf{k}}) - \delta(k^0 + E_{\mathbf{k}})\right], \tag{13}$$

and therefore

$$\widetilde{G}(k) = \sum_{s=\pm 1} \frac{\pi\, \delta(k^0 - sE_{\mathbf{k}})}{2E_{\mathbf{k}} \sinh(\beta E_{\mathbf{k}}/2)}. \tag{14}$$

5 Self-Energy and Width

The leading large-time ladder problem requires dressed side rails. Near the quasiparticle pole use

$$G_{\mathrm{R},a}(k) \simeq \frac{\mathrm{i}Z_a(\mathbf{k})}{2E_a(\mathbf{k})}\left[\frac{1}{k^0 - E_a(\mathbf{k}) + \mathrm{i}\Gamma_a(\mathbf{k})} - \frac{1}{k^0 + E_a(\mathbf{k}) + \mathrm{i}\Gamma_a(\mathbf{k})}\right]. \tag{15}$$

The width is obtained from the imaginary part of the on-shell retarded self-energy,

$$\Gamma_a(\mathbf{p}) = -\frac{Z_a(\mathbf{p})}{2E_a(\mathbf{p})}\,\mathrm{Im}\,\Pi_{\mathrm{R},a}(E_a(\mathbf{p}) + \mathrm{i}0, \mathbf{p}). \tag{16}$$

If a field is massless at tree level, the pipeline must first compute the thermal mass from the static self-energy:

$$m^2_{\mathrm{eff},a} = m^2_{0,a} + \Pi_{\mathrm{th},a}(0, \mathbf{0}). \tag{17}$$

For the Stanford scalar matrix benchmark,

$$m^2_{\mathrm{th}} = \frac{2\lambda}{3\beta^2}. \tag{18}$$

6 Rung Construction

The leading growth kernel is built from rung diagrams: two SK vertices, one on each real-time fold, joined by Wightman lines. Abstractly, for external on-shell rail species A, B, the rung has the form

$$R_{AB}(q) = \sum_{\gamma \in \mathcal{R}_{AB}} \frac{\mathcal{S}_\gamma}{|\,\mathrm{Aut}\,\gamma|} \int \prod_{\ell \in I_\gamma} \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,(2\pi)^d \delta^{(d)}\left(q + \sum_{\ell \in I_\gamma} \sigma_\ell \ell\right) \mathcal{I}_\gamma(q, \{\ell\}). \tag{19}$$

The integrand is

$$\mathcal{I}_\gamma = V^{L,\gamma}_{A;\{\alpha_\ell\}} V^{R,\gamma}_{B;\{\beta_\ell\}} \mathcal{C}_\gamma \prod_{\ell \in I_\gamma} \widetilde{G}_{\alpha_\ell \beta_\ell}(\ell), \tag{20}$$

where $\mathcal{C}_\gamma$ is the index/color contraction factor and $\mathcal{S}_\gamma$ includes statistics signs and multiplicities. The graph enumerator must retain only rungs whose large-N and weak-coupling order can contribute to the requested leading limit.

7 Ladder Equation

Let $f_A(\omega, p)$ denote the amputated ladder amplitude with rail species A. The exact ladder recursion has the schematic form

$$f_A(\omega, p) = -G_{\mathrm{R},A}(p) G_{\mathrm{R},A}(\omega - p)\left[h_A(p) + \sum_B \int \frac{\mathrm{d}^d k}{(2\pi)^d} R_{AB}(k - p) f_B(\omega, k)\right], \tag{21}$$

where h_A is the zero-rung source. The large-time growth exponent is independent of h_A, so the pipeline drops the inhomogeneous term and keeps the homogeneous equation.

Using the pole-pair replacement

$$G_{\mathrm{R},A}(p) G_{\mathrm{R},A}(\omega - p) \longrightarrow -\frac{\pi \mathrm{i} Z_A(\mathbf{p})^2}{2E_A(\mathbf{p})^2} \frac{\delta(p^0 - E_A(\mathbf{p})) + \delta(p^0 + E_A(\mathbf{p}))}{\omega + 2\mathrm{i}\Gamma_A(\mathbf{p})}, \tag{22}$$

the amplitude becomes on-shell supported. Write

$$f_A(\omega, p) = f_A(\omega, \mathbf{p})\,\delta\big((p^0)^2 - E_A(\mathbf{p})^2\big). \tag{23}$$

Then the real-time evolution equation is

$$\partial_t f_A(t, \mathbf{p}) = \sum_B \int \frac{\mathrm{d}^{d-1}\mathbf{k}}{(2\pi)^{d-1}} m_{AB}(\mathbf{k}, \mathbf{p}) f_B(t, \mathbf{k}) - 2\Gamma_A(\mathbf{p}) f_A(t, \mathbf{p}). \tag{24}$$

The on-shell gain kernel is

$$m_{AB}(\mathbf{k}, \mathbf{p}) = \sum_{\eta=\pm} \frac{Z_A(\mathbf{p}) Z_B(\mathbf{k}) R_{AB}(k_\eta - p)}{4E_A(\mathbf{p}) E_B(\mathbf{k})}, \qquad k_\eta - p = (E_B(\mathbf{k}) + \eta E_A(\mathbf{p}), \mathbf{k} - \mathbf{p}), \tag{25}$$

with the precise sign sum fixed by the pole assignments and statistics.

The operator implemented numerically is

$$(Mf)_A(\mathbf{p}) = \sum_B \int \frac{\mathrm{d}^{d-1}\mathbf{k}}{(2\pi)^{d-1}} m_{AB}(\mathbf{k}, \mathbf{p}) f_B(\mathbf{k}) - 2\Gamma_A(\mathbf{p}) f_A(\mathbf{p}). \tag{26}$$

Finally,

$$\lambda_L = \max\{\operatorname{Re}\mu : \mu \in \operatorname{spec} M\}. \tag{27}$$

8 Stanford Φ^4 Benchmark

The benchmark theory is a Hermitian matrix scalar in four spacetime dimensions, with

$$\mathcal{L} = \frac{1}{2} \operatorname{Tr}\left(\dot{\Phi}^2 - (\nabla\Phi)^2 - m^2\Phi^2 - 2g^2\Phi^4\right), \qquad \lambda = g^2 N. \tag{28}$$

For this theory, the leading rung is

$$R(q) = 48\lambda^2 \int \frac{\mathrm{d}^4\ell}{(2\pi)^4} \widetilde{G}(q/2 + \ell)\widetilde{G}(q/2 - \ell). \tag{29}$$

The corresponding on-shell kernel is

$$m(\mathbf{k}, \mathbf{p}) = \frac{R(k_+) + R(k_-)}{4E_{\mathbf{k}} E_{\mathbf{p}}}, \qquad k_\pm = (E_{\mathbf{k}} \pm E_{\mathbf{p}}, \mathbf{k} - \mathbf{p}). \tag{30}$$

The width can be eliminated in favor of the same rung:

$$\Gamma_{\mathbf{p}} = \frac{1}{6} \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{\sinh(\beta E_{\mathbf{p}}/2)}{\sinh(\beta E_{\mathbf{k}}/2)} m(\mathbf{k}, \mathbf{p}). \tag{31}$$

Therefore the eigenvalue equation is

$$\lambda_L f(\mathbf{p}) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} m(\mathbf{k}, \mathbf{p}) \left[f(\mathbf{k}) - \frac{\sinh(\beta E_{\mathbf{p}}/2)}{3\sinh(\beta E_{\mathbf{k}}/2)} f(\mathbf{p})\right]. \tag{32}$$

The closed form rung used for validation is

$$R(q) = \frac{6\lambda^2}{\pi\beta|\mathbf{q}|\sinh(|q^0|\beta/2)} \left[\Theta(-q^2 - 4m^2) \log\frac{\sinh x_+}{\sinh x_-} + \Theta(q^2) \log\frac{1 - e^{-2x_+}}{1 - e^{2x_-}}\right], \tag{33}$$

where

$$q^2 = -(q^0)^2 + |\mathbf{q}|^2, \qquad x_\pm = \frac{\beta}{4}\left[|q^0| \pm |\mathbf{q}|\sqrt{1 + \frac{4m^2}{|\mathbf{q}|^2 - (q^0)^2}}\right]. \tag{34}$$

9 One-Dimensional Reduction and Discretization

For rotationally invariant scalar models, set $P = |\mathbf{p}|$, $K = |\mathbf{k}|$, $E_P = \sqrt{P^2 + m^2}$. The angular integration uses

$$\int \mathrm{d}^3\mathbf{k} = 2\pi \int_0^\infty K^2\,\mathrm{d}K \int_{|K-P|}^{K+P} \frac{y\,\mathrm{d}y}{KP}, \qquad y = |\mathbf{k} - \mathbf{p}|. \tag{35}$$

The Stanford equation becomes

$$\lambda_L f(P) = \int_0^\infty \mathrm{d}K\, m_1(P,K) \left[f(K) - \frac{\sinh(\beta E_P/2)}{3\sinh(\beta E_K/2)} f(P) \right], \tag{36}$$

where $m_1(P,K)$ is the angle-averaged kernel obtained from Eq. (30) and Eq. (35). In the scalar benchmark the explicit expression is

$$m_1(P,K) = \frac{3\lambda^2}{(2\pi)^3\beta} \frac{K}{PE_PE_K} \int_{|K-P|}^{K+P} \mathrm{d}y\, \mathcal{B}(P,K,y), \tag{37}$$

with

$$\mathcal{B}(P,K,y) = \frac{\log\left[\sinh x_+^+/\sinh x_-^+\right]}{\sinh(\beta E_+/2)} + \frac{\log\left[(1-e^{-2x_+^-})/(1-e^{2x_-^-})\right]}{\sinh(\beta E_-/2)}, \tag{38}$$

$$E_\pm = |E_K \pm E_P|, \tag{39}$$

$$x_\pm^+ = \frac{\beta}{4}\left[E_+ \pm y\sqrt{1 + \frac{4m^2}{y^2 - E_+^2}}\right], \tag{40}$$

$$x_\pm^- = \frac{\beta}{4}\left[E_- \pm y\sqrt{1 + \frac{4m^2}{y^2 - E_-^2}}\right]. \tag{41}$$

The apparent $E_- = 0$ singularity must be handled by an analytic limit, not by an arbitrary regulator.

Map the half-line to the unit interval:

$$P(u) = P_0 \frac{u}{1-u}, \qquad 0 < u < 1. \tag{42}$$

Use $P_0 \simeq 3m$ in the small-mass benchmark and $P_0 \simeq m$ away from that regime. Define

$$f_2(u) = (1-u)^{-1} P(u) f(P(u)), \tag{43}$$

$$m_2(u,v) = P_0 \frac{(1-u)(1-v)}{P(u)K(v)} m_1(P(u), K(v)), \tag{44}$$

and

$$D(u) = \frac{P(u)}{1-u} \sinh \frac{\beta E_{P(u)}}{2}. \tag{45}$$

Then

$$\lambda_L f_2(u) = \int_0^1 \mathrm{d}v\, m_2(u,v) \left[f_2(v) - \frac{D(v)}{3D(u)} f_2(u) \right]. \tag{46}$$

After quadrature with nodes u_i and weights w_i, assemble

$$M_{ij} = w_j m_2(u_i,u_j) - \delta_{ij} \sum_\ell w_\ell m_2(u_i,u_\ell) \frac{D(u_\ell)}{3D(u_i)}. \tag{47}$$

The largest positive eigenvalue of M is the numerical estimate of λ_L. The code should symmetrize only after applying the mathematically correct similarity/weight transform; otherwise it should explicitly report the antisymmetric residual.

10 Leading-Behavior Extraction

The extractor should keep the symbolic prefactor of each kernel contribution. For a weak-coupling family of kernels,

$$M = \sum_{\gamma} c_\gamma \beta^{-1} \prod_{\alpha} (g_\alpha \beta^{\Delta_\alpha})^{n_{\gamma\alpha}} (\beta m_{\text{eff}})^{-r_\gamma} \widehat{M}_\gamma(\beta m_{\text{eff}}, \text{dimensionless ratios}), \tag{48}$$

the leading term is selected by the requested asymptotic ordering of g_α, $1/N$, and βm_{eff}. Numerically, this is checked by log–log fits:

$$\nu_g = \frac{\partial \log \lambda_L}{\partial \log g}, \qquad \nu_m = \frac{\partial \log \lambda_L}{\partial \log m_{\text{eff}}}, \qquad \nu_\beta = \frac{\partial \log \lambda_L}{\partial \log \beta}. \tag{49}$$

For the Stanford benchmark,

$$\lambda_L = c_m \frac{\lambda^2}{\beta^2 m} + \cdots, \qquad c_m \simeq 0.025, \qquad m\beta \ll 1,\ m \neq 0. \tag{50}$$

For the massless theory,

$$m_{\text{eff}} = m_{\text{th}} = \sqrt{\frac{2\lambda}{3\beta^2}}, \qquad \lambda_L = c_m \sqrt{\frac{3}{2}} \frac{\lambda^{3/2}}{\beta} \simeq 0.031 \frac{\lambda^{3/2}}{\beta}. \tag{51}$$

11 Implementation Algorithm

1. Parse the model into the data of Eq. (2).
2. Build S_{SK} by Eq. (4); verify Eq. (6) and the odd-a selection rule.
3. Derive SK vertices from Eq. (7); store branch coefficients separately from couplings, signs, and index tensors.
4. Derive G_{R}, ρ, and $\widetilde{G}$ from Eqs. (11)–(12).
5. Compute required thermal masses and widths using Eqs. (17) and (16).
6. Enumerate leading rung graphs and construct R_{AB} from Eq. (19).
7. Put rails on shell and assemble M using Eqs. (25)–(26).
8. Reduce by symmetries: rotational invariance, species blocks, charge sectors, and large-N planar sectors.
9. Discretize the resulting integral equation. For the scalar benchmark, use Eqs. (42)–(47).
10. Extract λ_L, eigenvectors, scaling exponents, and benchmark residuals.

12 Required Checks

The implementation should fail closed if any of these checks fails:

- $S_{\text{SK}}[\phi_a = 0] = 0$.
- No interaction vertex has an even number of a-legs.
- KMS relation: $\widetilde{G}(k) = \rho(k)/(2\sinh(\beta k^0/2))$ for bosons.

- Rung support and theta-function regions agree with the on-shell kinematics.
- Widths satisfy $\Gamma_a(\mathbf{p}) \geq 0$.
- The discretized operator reproduces the Stanford coefficients 0.025 and 0.031 within stated numerical tolerance.
- Leading exponents from symbolic prefactors agree with numerical log–log fits.

References

[1] Douglas Stanford, *Many-body chaos at weak coupling*, arXiv:1512.07687, https://arxiv.org/abs/1512.07687.